

TradingAI Strategies and High-Probability Trade Selection

Every current setup strategy, option-expression strategy and portfolio hedge - how each contributes to trade quality and expectancy.

Current system reference: 2026-08-29. Grounded in the current StrategyCatalog, Institutional Options strategy generator, strategy scoring/expected-move/expiration engines, M78 Setup Intelligence and portfolio-risk/allocation logic.

A named strategy is not inherently high probability. TradingAI narrows toward stronger trades by requiring alignment across setup, regime, volatility, expected move, contract quality, valuation, portfolio fit, execution and governance. M78 setup probabilities remain shadow-only until evidence matures and prospective certification is earned.

1. Three strategy layers

Layer	Examples	Purpose
Underlying setup	TREND_PULLBACK, BREAKOUT_CONTINUATION	Defines market mechanism and invalidation.
Option expression	LONG_CALL, BULL_CALL_SPREAD, IRON_CONDOR	Defines payoff, Greeks, theta/vega and capital.
Portfolio hedge	BUY_PROTECTIVE_PUT_SPREAD	Reduces portfolio risk rather than creating alpha.

2. The narrowing funnel

- Fresh canonical data
- Market regime/breadth/sector/dealer/volatility context
- Multi-timeframe Trend and Stock Intelligence
- M78 exact setup identity
- Regime-aware strategy eligibility
- Expected-move and volatility fit
- Expiration and strike optimization
- Greeks and liquidity
- Option valuation and relative value
- Institutional Decision Intelligence
- Portfolio fit and capital allocation
- Trade Builder revalidation
- Executable-price/broker preflight
- Dynamic management and expiration governance

3. Strategy scoring dimensions

Dimension	Value
Technical/thesis	Prevents direction mismatch.
Volatility	Chooses whether premium is attractive to buy/sell.
Expected move	Matches long-gamma vs bounded premium structures.
Strike	Controls delta, break-even and payoff.
Expiration	Matches horizon and theta/event exposure.
Greeks	Verifies intended sensitivities.
Liquidity	Protects edge from spreads/slippage.
Execution	Tests fillability.
Risk/reward	Rejects poor asymmetry.
Data confidence	Downweights weak/stale evidence.
Portfolio fit	Prevents concentration/correlation damage.

4. M78 underlying setup strategies

These define the market mechanism. They do not automatically place a trade.

TREND_PULLBACK

Definition: Retracement inside intact trend

How it adds value: Improves entry and stop geometry versus chasing

Typical option expression: Long call/put or debit spread; credit spread if IV rich

TREND_CONTINUATION

Definition: Persistent established trend

How it adds value: Captures sustained directional edge with alignment

Typical option expression: Long option/debit spread; credit spread for steady high-IV trend

BREAKOUT_SETUP

Definition: Pressure near resistance before confirmation

How it adds value: Watch state avoids premature breakout entries

Typical option expression: Wait/conditional until confirmation

BREAKOUT_CONFIRMED

Definition: Resistance broken with confirmation

How it adds value: Filters many false breaks

Typical option expression: Long call or bull call spread

BREAKOUT_RETEST

Definition: Breakout tests former resistance/new support

How it adds value: Confirmation plus improved entry/invalidation

Typical option expression: Long call/bull call spread; calendar/diagonal if slower

BREAKOUT_CONTINUATION

Definition: Upside resumes after break/retest

How it adds value: Structural validation plus persistence

Typical option expression: Long call/bull call spread; bull put spread if IV rich

BREAKDOWN_SETUP

Definition: Pressure near support before confirmation

How it adds value: Watch state avoids assuming support failure

Typical option expression: Wait/conditional until confirmation

BREAKDOWN_CONFIRMED

Definition: Support broken with confirmation

How it adds value: Filters false breakdowns

Typical option expression: Long put or bear put spread

BREAKDOWN_RETEST

Definition: Breakdown tests former support/new resistance

How it adds value: Better bearish entry than chasing first selloff

Typical option expression: Long put/bear put spread; put calendar/diagonal

BREAKDOWN_CONTINUATION

Definition: Downside resumes after break/retest

How it adds value: Validation plus bearish persistence

Typical option expression: Long put/bear put spread; bear call spread if IV rich

FAILED_BREAKOUT_REVERSAL

Definition: Breakout fails back below resistance

How it adds value: Captures trapped-long reversal mechanism

Typical option expression: Long put/bear put spread; bear call spread

FAILED_BREAKDOWN_REVERSAL

Definition: Breakdown fails and reclaims support

How it adds value: Captures trapped-short reversal

Typical option expression: Long call/bull call spread; bull put spread

SUPPORT_REVERSAL

Definition: Price rejects governed support

How it adds value: Nearby invalidation improves bullish asymmetry

Typical option expression: Long call/bull call spread; bull put spread

RESISTANCE_REVERSAL

Definition: Price rejects governed resistance

How it adds value: Nearby invalidation improves bearish asymmetry

Typical option expression: Long put/bear put spread; bear call spread

5. Canonical executable option strategies

The current StrategyCatalog supports these 14 strategies. Institutional Options uses equivalent aliases such as CALL_CALEDAR/CALL_DIAGONAL and PUT_CALEDAR/PUT_DIAGONAL.

LONG_CALL

Structure: Bullish long premium

Best fit: Bullish expansion; low-moderate IV

How it adds value: Open-ended convex upside with defined premium risk

Main failure mode: Theta/IV overpayment

LONG_PUT

Structure: Bearish long premium

Best fit: Bearish expansion; low-moderate IV

How it adds value: Convex downside with defined premium risk

Main failure mode: Rich downside IV without movement

BULL_CALL_SPREAD

Structure: Bullish debit vertical

Best fit: Bullish trend; moderate-high IV

How it adds value: Reduces volatility/theta cost and capital

Main failure mode: Caps upside

BEAR_PUT_SPREAD

Structure: Bearish debit vertical

Best fit: Bearish trend; moderate-high IV

How it adds value: Efficient defined-risk bearish exposure

Main failure mode: Caps crash payoff

BULL_PUT_SPREAD

Structure: Bullish credit vertical

Best fit: High IV; strong support; positive gamma

How it adds value: Can profit on rise/flat/modest decline

Main failure mode: Gap/tail risk

BEAR_CALL_SPREAD

Structure: Bearish credit vertical

Best fit: High IV; strong resistance; positive gamma

How it adds value: Can profit on decline/flat/modest rise

Main failure mode: Breakout risk

IRON_CONDOR

Structure: Defined-risk range structure

Best fit: Range; high IV; positive gamma

How it adds value: Monetizes containment and rich premium

Main failure mode: Directional escape

IRON_BUTTERFLY

Structure: Concentrated range structure

Best fit: Strong center; high IV; compression

How it adds value: Higher premium around mean-reversion center

Main failure mode: Narrow profitable region

LONG_STRADDLE

Structure: Long call + put near same strike

Best fit: Large move; direction uncertain; move underpriced

How it adds value: Trades magnitude rather than direction

Main failure mode: Two premiums/theta

LONG_STRANGLE

Structure: OTM long call + put

Best fit: Very large move; direction uncertain

How it adds value: Cheaper two-sided convexity

Main failure mode: Wider break-evens

CALENDAR_CALL

Structure: Call time spread

Best fit: Bullish/neutral transition; favorable term structure

How it adds value: Slower thesis plus front-decay/term-structure edge

Main failure mode: Complex event/multi-expiry risk

CALENDAR_PUT

Structure: Put time spread

Best fit: Bearish/neutral transition; favorable term structure

How it adds value: Slower bearish thesis plus term-structure edge

Main failure mode: Complex event/multi-expiry risk

DIAGONAL_CALL

Structure: Different call strikes/expiries

Best fit: Bullish mature trend/transition

How it adds value: Directional exposure plus front-leg decay harvest

Main failure mode: Complex Greeks/expiry management

DIAGONAL_PUT

Structure: Different put strikes/expiries

Best fit: Bearish mature trend/transition

How it adds value: Bearish exposure plus front-leg decay harvest

Main failure mode: Complex Greeks/expiry management

6. Long-premium versus premium-selling regime logic

Environment	Preference	Rationale
Implied move underpriced / high move	Long premium / directional / long gamma	Pay for convexity when realized movement can exceed pricing.
Implied move overpriced / low-normal move	Defined-risk premium selling	Collect rich premium when containment is credible.
Low/normal volatility	Long calls/puts/straddles/strangles	Avoid overpaying for volatility.
High/extreme volatility	Credit spreads/condors/butterflies	Harvest richer premium with defined risk.
Negative gamma / expansion	Long call/put or debit spread	Market structure can amplify movement.
Positive gamma / containment	Credit spreads / condor / butterfly	Dealer hedging can damp movement.

7. Expiration fit used by the current engine

Family	Highest current DTE score
LONG_CALL / LONG_PUT	21-45 DTE
BULL_CALL / BEAR_PUT	21-60 DTE
BULL_PUT / BEAR_CALL	21-45 DTE
IRON_CONDOR / IRON_BUTTERFLY	21-45 DTE
LONG_STRADDLE / LONG_STRANGLE	14-45 DTE
Calendar / diagonal	Multiple expiries by design

8. Portfolio hedge strategy

BUY_PROTECTIVE_PUT_SPREAD

Portfolio allocation can recommend a protective SPY put spread when beta-weighted directional exposure is excessive. It targets a reduction in portfolio beta-weighted delta. Its value is risk reduction, not standalone alpha.

The codebase also references delta, directional, volatility and sector-downside hedge intents. These are risk-management intents and are not all canonical StrategyCatalog structures.

9. Aliases and referenced non-canonical labels

Label	Interpretation
CALL_DEBIT_SPREAD / CALL_SPREAD / PUT_SPREAD	Generic or legacy vertical-spread labels; canonical catalog uses explicit bull/bear names.
CALL_CALENDAR / PUT_CALENDAR	Institutional aliases for CALENDAR_CALL / CALENDAR_PUT.
CALL_DIAGONAL / PUT_DIAGONAL	Institutional aliases for DIAGONAL_CALL / DIAGONAL_PUT.
SHORT_CALL / SHORT_PUT	Referenced by scoring logic; naked shorts are not in the canonical StrategyCatalog.
CALL_RATIO_SPREAD / PUT_RATIO_SPREAD	Referenced labels, not canonical StrategyCatalog entries.
PROTECTIVE_PUT	Risk concept; current portfolio recommendation uses BUY_PROTECTIVE_PUT_SPREAD.

10. How combinations narrow the strongest trades

Underlying evidence	Volatility / move	Likely expression	Selection benefit
Bull trend pullback	IV low-normal; expansion plausible	LONG_CALL / BULL_CALL_SPREAD	Trend + improved entry + convex/defined-risk upside.
Bull pullback with strong support	IV high; contained move	BULL_PUT_SPREAD	Support + premium richness creates wider favorable region.
Breakout continuation	Negative gamma; move underpriced	LONG_CALL / BULL_CALL_SPREAD	Structural confirmation + acceleration potential.
Failed breakout reversal	High IV; resistance strong	BEAR_CALL_SPREAD / BEAR_PUT_SPREAD	Trap mechanism + structural resistance + defined risk.
Range/compression	Positive gamma; IV high	IRON_CONDOR / IRON_BUTTERFLY	Containment + rich premium + theta.
Large event move, direction uncertain	Move underpriced	LONG_STRADDLE / LONG_STRANGLE	Trades magnitude instead of forcing direction.
Slow bullish transition	Term structure attractive	CALENDAR_CALL / DIAGONAL_CALL	Matches slower path and term-structure edge.

11. Why the platform does not simply choose the highest win rate

- A high probability with poor payoff can have negative expected value.
- Short-premium strategies can show high nominal win rates while carrying rare large losses.
- Long-premium strategies can win less often but produce convex payoffs when the distribution is favorable.
- Liquidity/slippage can erase theoretical probability/EV.
- A good standalone trade can worsen portfolio concentration or tail risk.
- Probability must be calibrated on point-in-time outcomes; M78 refuses training when sample gates are not met.
- Therefore the platform combines probability, payoff, valuation, portfolio fit and execution rather than optimizing win rate alone.

12. Current governance and evidence status

M78 successfully captures explicit setup populations, but the observed current state still has insufficient matured setup outcomes for model training. No active M78 shadow model exists. The production system therefore continues to rely on its certified production intelligence/decision/portfolio authorities while M78 accumulates prospective evidence.

The practical goal is not to manufacture a 'highest probability' label. It is to progressively remove trades whose mechanism, pricing, portfolio impact or execution quality do not justify risk, leaving a smaller set with the strongest explainable expected-value case.